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Nurturing hopes: Progress in implementing the Global Initiative for Childhood Cancer in the WHO South-East Asia Region

World Health Organization
South-East Asia Region

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Abbreviations and acronyms

childGICR	expansion of Global Initiative for Cancer Registry Development in 2020 to include childhood cancers
GICC	Global Initiative for Childhood Cancer
GICR	Global Initiative for Cancer Registry Development
GLOBOCAN	Global Cancer Observatory
GPACCM	Global Platform for Access to Childhood Cancer Medicines
HICs	high-income countries
imPACT	Integrated Mission on Programme of Action for Cancer Therapy
IARC	International Agency for Research on Cancer
LMICs	low- and middle-income countries
NCCP	National Cancer Control Plan
MoH	Ministry of Health
NCD	noncommunicable disease
PBCR	population-based cancer registry
PHC	primary health care
SDG	Sustainable Development Goal
SEACanGrid	South-East Asia Cancer Grid
SEAR-CCN	South-East Asia Regional Childhood Cancer Network
St. Jude	St. Jude Children's Research Hospital
UHC	universal health coverage
UNICEF	United Nations Children's Fund
VTB	Virtual Tumour Board
WHO	World Health Organization
WHO HQ	WHO headquarters
WHO-SEARO	WHO Regional Office for South-East Asia

Foreword

The Global Initiative for Childhood Cancer (GICC) was launched by WHO in 2018, alongside partners including St. Jude Children's Research Hospital, with a worldwide commitment to achieve a global survival rate of 60% for childhood cancer and to reduce suffering for all by 2030.

In the WHO South-East Asia Region, childhood cancer remains a significant public health challenge, where disparities in access to timely diagnosis and quality treatment have historically led to lower survival rates compared to high-income settings.

This publication offers an overview of the steps taken collectively by our Member States, partners, and health professionals since the start of GICC implementation in the region.

The establishment of the South-East Asia Regional Childhood Cancer Network (SEAR-CCN) has been significant. It has created a platform for sharing expertise, building workforce capacity, and standardizing care through initiatives such as the virtual tumour board and external quality assurance for pathology services. The Global Platform for Access to Childhood Cancer Medicines—which aims to provide an uninterrupted supply of quality-assured cancer medicines to low- and middle-income countries (LMICs)—offers renewed hope for equitable access to quality assured cancer medicines.

The progress in the GICC focus countries shows that evidence-based interventions can be tailored to country needs and be scaled up. This includes the building of resilient shared care networks in Myanmar, strong advocacy-driven advancements in Nepal, and the strategic planning demonstrated by Sri Lanka.

I extend my appreciation and thanks to the individuals and institutions that have contributed to this progress. Even as we celebrate these successes, we must remain aware that our journey is not over. I encourage all stakeholders to use the insights in this report to renew and accelerate efforts to expand workforce capacity, improve quality of care, and improve equitable access.

Together, with continued collaboration and commitment, we can fulfill the promise of GICC and move toward a future where every child with cancer has a greater chance of survival and a full, healthy life.

Dr Catharina Boehme
Officer-in-Charge
WHO South-East Asia

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Leadership and coordination

This report has been drafted by NCD team members at the WHO Regional Office for South-East Asia (WHO-SEARO) – Dr Bishnu Rath Giri and Dr Nalika Gunawardena. Dr Cherian Varghese, former Director a.i. at Healthier Population and Noncommunicable Diseases (HPN) Department in WHO-SEARO, has contributed to the conception and outline of this report. Technical oversight and guidance were provided by Mr Hussain Rasheed, Director, HPN.

WHO contributors and reviewers

The cancer focal points at WHO country offices – including Dr Thida Moe, Dr Suman Panthi, Dr Gampo Dorji, Dr Farrukh Qureshi, and WHO headquarters (WHO HQ) cancer team members Dr Roberta Ortiz and Dr Andr Ilbawi – have contributed to this work.

Institutional contributions

WHO acknowledges the remarkable work on childhood cancer and inputs for this publication from focal points in the ministries of health in the Global Initiative for Childhood Cancer (GICC) focus countries and South-East Asia Regional Childhood Cancer Network (SEAR-CCN) member institutions, including Dr Aye Aye Khaing, Dr Sudhir Sapkota, Dr Sucharita Tuladhar, Dr Sani Sipai, Dr Sanjeeva Gunasekera, Dr Suraj Perera, Dr Girish Chinnaswamy, Dr Maya Prasad, Dr Rachna Seth, Dr Arpita Bhattacharya, Dr Soehartati Gondhowiardjo and Dr Mayang Permata.

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Executive summary

About 56 000 children were newly diagnosed with cancer in the WHO South-East (SE) Asia Region in 2022. The rate of mortality in childhood cancer was over 45%, more than three times higher than that for high-income countries (HICs). WHO launched the Global Initiative for Childhood Cancer (GICC) in 2018, with the objective of increasing prioritization of childhood cancer at the global, regional and national levels. The overall goal of the GICC is to achieve a 60% global survival rate by 2030. The initiative has developed a strategic approach – the **Cure*All*** framework – consisting of four pillars and three enablers to help increase the capacity of countries to provide quality services for children with cancer.

An assessment at the start of the activities under the GICC in the Region identified gaps on various fronts in childhood cancer care, including policy, quality of diagnosis and treatment, and access to these services. To harness the collective strength of countries to address the gaps, the WHO Regional Office for South-East Asia (WHO-SEARO) established the South-East Asia Regional Childhood Cancer Network (SEAR-CCN) in 2020 to serve as a platform for key institutions for collaboration in technical capacity-building and patient care.

Since its establishment, the SEAR-CCN has been carrying out activities to support countries and institutions in the Region for improving childhood cancer services. The regional Virtual Tumor Board and pathology external quality assurance services, established for childhood cancers by the SEAR-CCN, promote evidence-based care improvement in childhood cancer and are being effectively utilized by institutions across the countries in the Region. To support building the capacity of nurses in paediatric oncology, the Network has established virtual and hands-on training, which had been attended by 170 nurses from 25 institutions, as of December 2024. SEAR-CCN institutions have jointly developed resource guides for shared care centres and training modules on common paediatric oncology procedures. The Network has facilitated bilateral collaborations between institutions across the countries, advancing shared patient care and strengthening human resource capacity for childhood cancer services.

While all countries in the SE Asia Region are being supported at various levels of engagement through the GICC, four countries have been designated as focus countries, based on demonstration of early progress with advances in one or more of the **Cure*All*** pillars and enablers, commitment to improving childhood cancer care, and formal recognition in writing between the governments and WHO. The GICC focus countries in the Region are Bhutan, Myanmar, Nepal and Sri Lanka. The focus countries have implemented innovative strategies to improve childhood cancer care that could be adopted in other countries.

Myanmar has established an extensive network of 41 satellite centres, easing the burden on families who would otherwise need to travel to distant comprehensive treatment facilities. Currently, about one third of newly diagnosed cases receive treatment from these satellite centres in Myanmar.

In Nepal, advocacy has been a major driver for change. A shared care centre network has been established, while national guidelines for management of Wilms tumor have been drafted. The country has also been selected to participate in the Global Platform for Access to Childhood Cancer Medicines (GPACCM), thus availing of quality assured medicines delivered free of cost to children with cancer.

Sri Lanka has developed a standalone strategic plan for childhood and adolescent cancers that has guided actions in multiple aspects of childhood cancer care. The establishment of the hospital-based childhood cancer registry is one of the notable achievements. In addition, for improving the quality of care at the National Cancer Institute, an antibiotic stewardship programme has been established and a pain-free procedure protocol developed. The country has also developed guidelines for pathological diagnosis of nephroblastoma and retinoblastoma. Preparations are underway for the country's participation in the GPACCM.

Bhutan has prioritized childhood cancer and is building the capacity of programme managers for including childhood cancer in national policies. Childhood cancer service providers have been participating in the capacity-building activities carried out by the SEAR-CCN and institutional collaborations are being strengthened for shared care for childhood cancer.

These initial successes have been enabled by effective advocacy generating political will, clearly defined governance structure linked to policy and leveraging financial means from domestic sources. However, there are challenges specific to individual countries. Hence, sustaining the achievements and accelerating the progress require persistent efforts on addressing the gaps in individual countries, including policy development, resource mobilization, capacity-building and expanding access to quality services by strengthening health systems.

1. An overview of childhood cancer

It is estimated that over 400 000 children and adolescents (aged 0–19 years) worldwide develop cancer every year. Though numerically this may seem small, in terms of disability-adjusted life years (DALY), childhood cancers together are estimated to rank as the sixth leading cause of cancer burden and ninth leading childhood disease burden.[1,2]

Cancer patterns in children differ from those among adults. Leukaemia is by far the most common cancer type in children. Other common ones include lymphomas, Wilms tumour of kidneys, retinoblastoma and central nervous system tumours. Together, these cancers constitute about half of all childhood cancer incidence.[1] As the organs and tissues are in stages of development in childhood, cancers in children behave differently. They generally grow rapidly and their patterns of spread through the body are also different. Because of the fast growth, children become critically ill in early cancer stages. This rapid growth also leads to distinct tumour biology, microscopic appearance and clinical behaviour, making them more sensitive and responsive to chemotherapy. Due to these factors, childhood cancers need a distinct strategy to diagnose, treat and manage.[3]

In high-income countries (HICs), more than 80% of the children with cancer survive with effective treatment through multimodality therapy, including optimized use of conventional chemotherapeutic agents. In low- and low-middle-income countries (LMICs), the survival rate is less than 30%.[4] This disparity is a result of multiple factors. A large proportion of children with cancer do not get diagnosed, while some get misdiagnosed. Among those getting a diagnosis, some families refuse treatment, while another significant proportion of children abandon treatment midway. There are issues concerning both access and suboptimal quality of services. The proportion of deaths due to toxicity and relapses is also higher in the low- and low-middle-income countries.[5]

The likelihood of a child with cancer surviving is dependent on the ability of his or her family to access essential cancer services without incurring financial burden. A significant proportion of childhood cancer deaths can be avoided through a public health approach to cancer control – one that prioritizes universal access to high-quality care, ensures financial protection, and supports continuous surveillance of disease patterns and treatment outcomes. Investing in a set of essential, comprehensive interventions can help achieve significant improvement in survival rates.[6] Addressing childhood cancer care gaps, therefore, presents one of the low hanging fruits in global public health. Reducing mortality in the 0–5 years age group and improving universal access to care for all children with cancer will also contribute to achieving of multiple targets related to good health and well-being under the Sustainable Development Goal (SDG) 3.[7]

The Global Initiative for Childhood Cancer

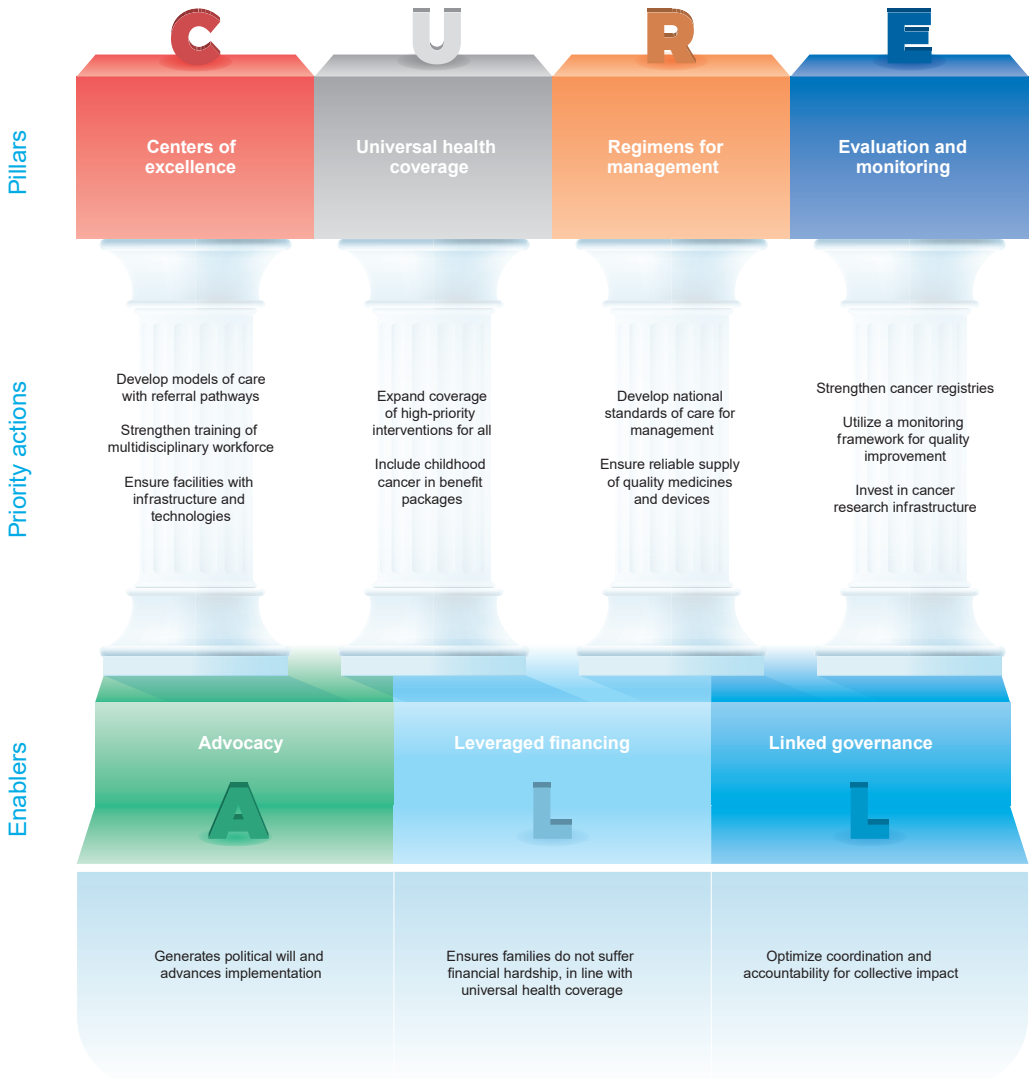
Keeping in view the need to reduce inequities and generate political commitment for the SDG agenda, WHO in 2018 launched the Global Initiative for Childhood Cancer (GICC).[8] The initiative, rolled out with support from the US-based St. Jude Children's Research Hospital (St. Jude) and other global partners, has set the goal of achieving at least 60% childhood cancer survival globally by 2030, seeking to

reduce suffering among all children with cancer. It is an ambitious goal that represents a doubling of the current global cure rate for children with cancer, saving an estimated additional one million lives.

Reaching the target requires focused interventions at multiple levels. These include strengthening health systems to enable primary health providers to reliably identify early signs and symptoms of childhood cancer and refer affected children to specialized treatment centres; adopting a universal health coverage (UHC) approach to ensure equitable access to services; strengthening comprehensive childhood cancer centres to provide standardized care; and making palliative care services available for children by integrating services with mainstream health delivery system.

The initiative has two primary objectives – increasing the capacity of countries to provide quality services for children with cancer and enhancing prioritization of childhood cancer at global, regional and national levels.[7] To translate the initiative's goals and objectives into interventions at national, regional and global levels, WHO has developed the **Cure*All*** framework, a strategic approach and a "how to" guide for policy-makers and programme managers. The framework embodies four key pillars and three enablers, outlining the coordinated operational approach needed to establish, scale up and sustain the National Childhood Cancer Programme and an approach to monitor progress (Fig. 1).[5]

Fig. 1. CureAll framework for implementation of the Global Initiative for Childhood Cancer



The four pillars of the **Cure*All*** framework are:

- **c**entres of excellence and care networks with a sufficient and competent workforce to increase the capacity to deliver quality, patient-centred services;
- **u**niversal health coverage by integrating childhood cancer as part of the full range of essential, quality-assured services and ensuring its inclusion in benefit packages;
- **r**egimens and roadmaps for diagnosis and treatment that are context-appropriate and facilitate delivery of quality services through evidence-based utilization of essential health products; and
- **e**valuation and monitoring, with robust information systems and research to ensure effective implementation, coupled with research on the unique genetic, epigenetic, environmental and societal aspects of childhood cancer in different populations.
- The **Cure*All*** framework identifies three enablers for successful implementation of the GICC:
- **a**dvocacy that generates political will and advances implementation;
- **l**everaged financing to ensure families do not suffer financial hardship, in line with universal health coverage; and
- **l**inked governance to optimize coordination and accountability for collective impact.

The GICC has identified six childhood cancers – acute lymphoblastic leukaemia, Burkitt lymphoma, Hodgkin lymphoma, retinoblastoma, Wilms tumour and low-grade glioma – as index cancers for monitoring the progress of the implementation of the **Cure*All*** framework in countries. These cancers have been selected because they represent the majority of incident cases, have affordable and effective therapies that yield high survival rates, and require multidisciplinary management through coordinated team-based approaches.[7]

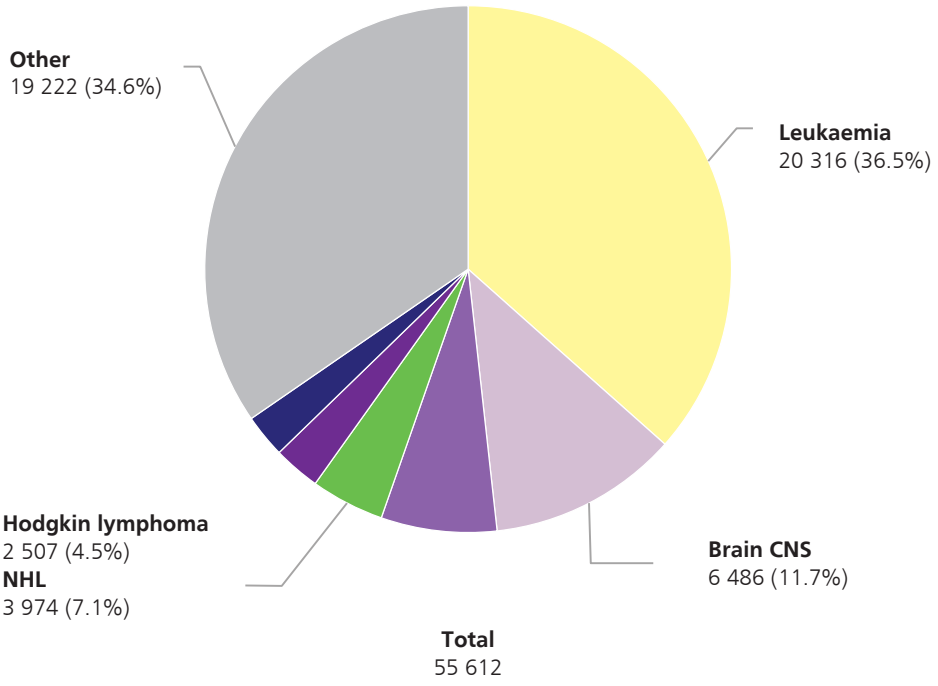
2. Status of childhood cancer in the WHO South-East Asia Region

Childhood cancer patterns

Based on the estimates of the Global Cancer Observatory (GLOBOCAN), about 56 000 new cases of childhood cancer were diagnosed in the WHO South-East (SE) Asia Region in the year 2022.[9] In 2022, the Region comprised 11 countries, namely Bangladesh, Bhutan, Democratic People's Republic of Korea (DPR Korea), India, Indonesia,* Maldives, Myanmar, Nepal, Sri Lanka, Thailand and Timor-Leste. A total of 56 000 new cases of childhood cancer represent the highest annual number of new childhood cancer cases among WHO regions. Based on the estimates on the burden by cancer types, leukaemia was the leading type, accounting for about 37% of the cases, followed by central nervous system tumours (11.7%) and non-Hodgkin lymphoma (7.1%) (Fig. 2).

The childhood cancer mortality rate in the Region is more than three times higher than that for high-income countries.[9] Over 45% of the children with cancer in the Region were estimated to have succumbed to the disease in 2022.[10]

Fig. 2. Distribution of cancers among children in South-East Asia Region



Source: Globocan 2022

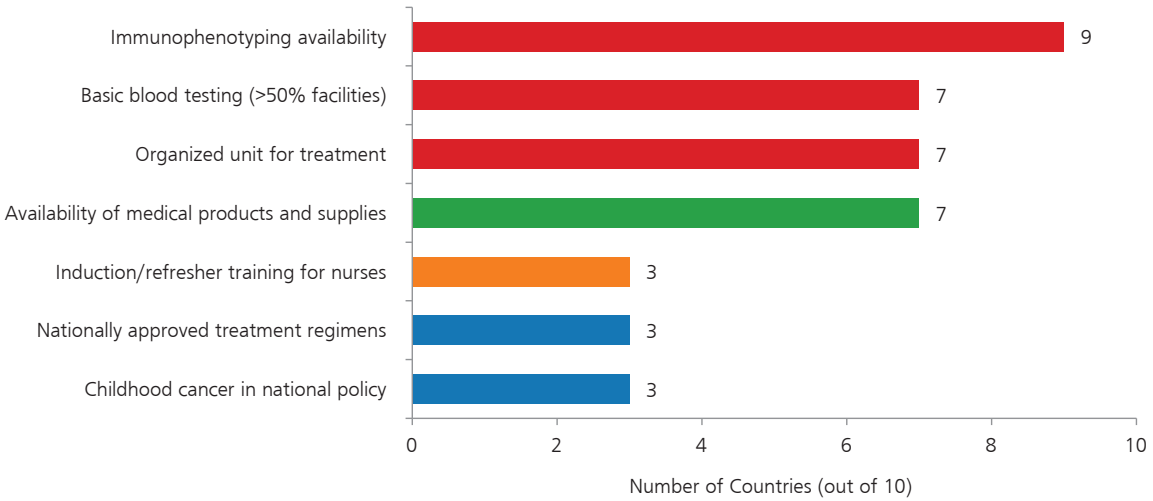
* Data from Indonesia are included as it was part of the South-East Asia Region at the time this report was commissioned and prepared. Indonesia was reassigned from the WHO SE Asia Region to the Western Pacific Region with effect from June 2025 vide a resolution to that effect promulgated by the Seventy-eighth World Health Assembly.

Capacity of health systems to manage childhood cancers

In 2021, the WHO Regional Office for South-East Asia (WHO-SEARO) conducted an assessment of national health systems' capacity to manage childhood cancers using a questionnaire. The focal points for childhood cancer in the ministries of health of all 11 countries were invited to participate, out of which 10 countries other than DPR Korea responded.[11] Among the 10 countries that responded to the survey, only three had childhood cancer specifically included in their national policy documents. Nationally approved treatment regimens for some of the common childhood cancers were available only in three countries. Seven countries reported having services for testing total leukocyte count, which is the basic blood test to indicate a possible diagnosis of leukaemia, at more than 50% of the primary health care facilities in the countries.

Nine countries reported the existence of centres offering immunophenotyping by flowcytometry for a definitive diagnosis of acute leukaemia. The availability of the service ranged from 0.02 to 1.84 per million population. Seven countries reported having an organized unit for treatment of childhood cancer and also indicated that more than 70% of the critical medicines and devices listed in the survey were "mostly available" (Annex 1). The nurse-to-bed ratio in the paediatric oncology units ranged widely from 1:4 to 1:34, with most countries having a higher number of beds per nurse during night shifts. This is noted to be far less than the recommended nurse-to-bed ratio of 1:5 for low- and middle-income countries.[12] Three countries reported having the provision of induction and refresher training for nurses on paediatric oncology nursing at more than 50% of the treatment centres (Fig. 3).

Fig. 3. Health System Capacity for Childhood Cancer Care – WHO SEARO Assessment (2021)



The Noncommunicable Disease (NCD) Country Capacity Survey (NCD CCS) is a global periodic assessment by WHO of national capacity for NCD prevention and control. Government officials are nominated to respond to the survey. In 2023, the survey made inquiries into the availability of services for early detection of childhood cancer as a service integrated into primary health care (PHC). Only two countries responded positively. Six countries reported that tertiary-level cancer centres dedicated to children were "generally available", reaching more than 50% of patients in need.

One country reported the existence of a population-based cancer registry (PBCR) with national coverage; four countries have hospital-based cancer registries (HBCRs) with national coverage; five countries have PBCRs with subnational coverage; and one country has an HBCR with subnational coverage. All cancer registries also cover childhood cancer.[13]

On analysing the cancer-related policy and plans of individual Member States in the Region, it is seen that, as of 2025, seven countries have included childhood cancer as a priority in their national plans, either as a standalone or as part of the general cancer or NCD plan.

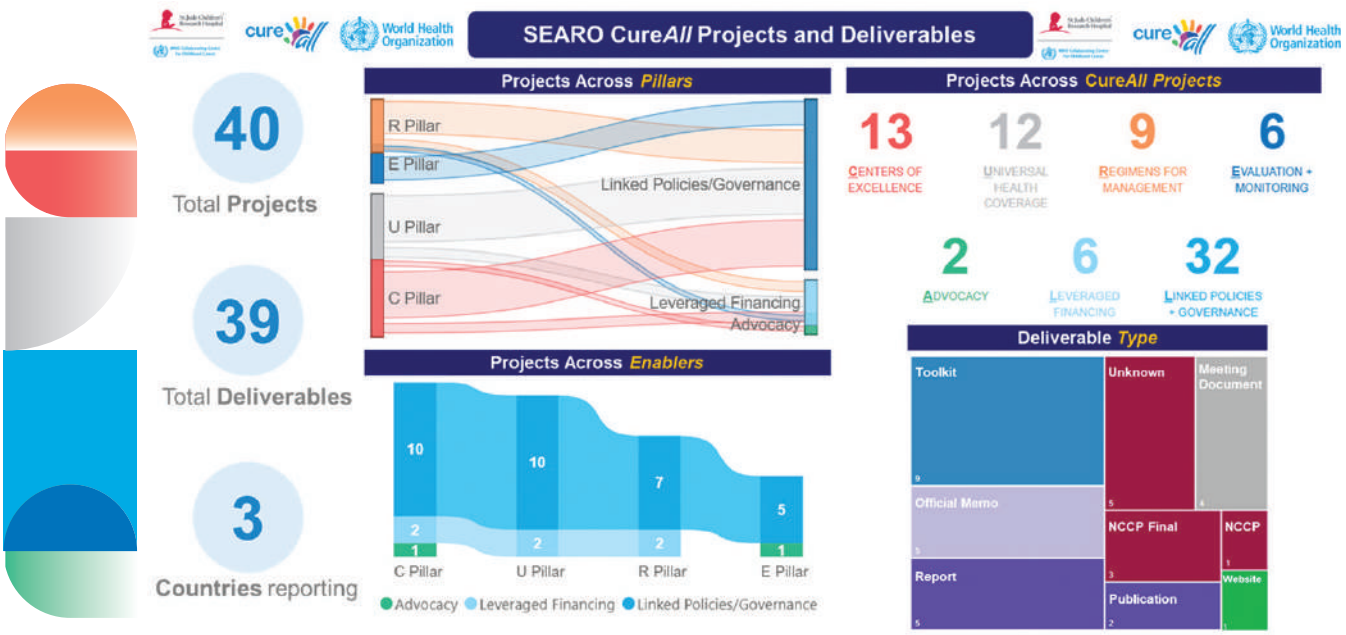
Palliative care is important for children with cancer during both curative intent treatment and end-of-life care to ensure comfort and quality of life, and to provide emotional and psychological support to affected families.[14] The Global Atlas for Palliative Care shows that most countries in the Region are in early stage of capacity-building for palliative care in general and palliative care for childhood cancers. Among all countries in the Region, only Thailand provides palliative care for children that is integrated into mainstream health-care services.[15]

3. Progress in GICC implementation in the WHO South-East Asia Region

A child with retinoblastoma under treatment at Yangon Children's Hospital, Myanmar

This report is a narrative synthesis of the progress of the implementation of the GICC in the WHO SE Asia Region from its initiation in 2020 to May 2025, thus reporting the collective progress of all 11 countries included in the Region during the period. An infographic summary of the GICC implementation activities till March 2025 in the Region is presented in the country progress dashboard below (Fig. 4).

Fig. 4. Cure*All* Country Progress Dashboard (2025)



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Establishing the South-East Asia Regional Childhood Cancer Network

The WHO SE Asia Region has focused on childhood cancer since 2020 through coordinated responses aligned with the **Cure*All*** framework. In July 2020, WHO-SEARO established the South-East Asia Childhood Cancer Network (SEAR-CCN) to guide the process.[16] WHO requested the ministries of health to nominate institutions to join the Network as members. The eligibility criteria for membership included those providing services for childhood cancer and with the potential to drive changes in childhood cancer in the country or the geo-administrative region they represented. Eleven institutions from six countries were included in the initial phase.

The SEAR-CCN aims to build a community of practice among institutions for implementing evidence-based interventions to improve the quality of and access to care and thus, catalyse the efforts by countries to enhance care for children with cancer. In addition to member institutes, the SEAR-CCN invited policy-makers and members from civil society to contribute to the technical dialogues, whenever relevant to specific activities of the Network.

Core action areas for the South-East Asia Regional Childhood Cancer Network

In alignment with the broader aims of the Network, the following core action areas were identified to be implemented in the countries:

- Initiate advocacy for policy change and reforms in childhood cancer service delivery.
- Promote implementation of evidence-based care improvement interventions.
- Build networks within the countries for childhood cancer to share resources and technical expertise for childhood cancer services.

- Build capacity and create training programmes on clinical care for diverse professionals in childhood cancer.
- Foster bilateral or multilateral collaborative research among Network members.
- Identify and deploy locally available resources for care improvement and widen coverage of childhood cancer services.

Assessment of the strengths and gaps of the member institutes of the South-East Asia Regional Childhood Cancer Network

For the initial years of operation of the Network, while gradually increasing the engagement in activities having broader, nationwide scope, the SEAR-CCN decided to sharpen the focus on activities related to building the capacity of the member institutes themselves so as to facilitate their evolution as centres of excellence for childhood cancer. This required identification of institutes within the SEAR-CCN, each with a specific set of strengths that could be utilized to support another institute. Hence, the Network collected institution-specific information through a questionnaire. Paediatric oncology leads from the 11 member institutions responded to this questionnaire.

This exercise showed that the bed capacity available at the institutions was proportionate to the patient load. Basic pathology, radiology and surgery facilities were available at all institutions. The exercise also revealed gaps in the institutions, specifically with regard to high-end diagnostics, care coordination, multidisciplinary care, quality measures, and opportunities for clinical and leadership trainings at the institutions.[17]

The activities of the SEAR-CCN on capacity-building, multidisciplinary care and quality improvement were designed based on the findings of the assessment. Member institutions expressed willingness to collaborate amongst themselves to explore capacity-building measures, including virtual and in-person training, observational visits and formal fellowship trainings. The need for capacity-building for nursing was prioritized and it was agreed that establishing a virtual training programme would be explored.

The SEAR-CCN also decided on setting up a Virtual Tumour Board to strengthen the multidisciplinary team approach to patient care. It was also noted that childhood cancer registries required improvement for evidence-based planning. Member institutions agreed to prioritize collaboration on strengthening the hospital-based cancer registry linked with a population-based cancer registry.

The SEAR-CCN became operational in August 2020 amidst the initial peak of COVID-19 in the Region; all activities were coordinated virtually since its inception. By 2024, the Network has grown to include 21 member institutions from 10 Member States (Annex 2). Together, these institutions have been providing improved care to over 7500 children newly diagnosed with cancer every year.[18] The subsequent sections describe some of the key activities of the Network that have contributed to improving childhood cancer care in the Region.

Activities of the South-East Asia Regional Childhood Cancer Network

Promoting evidence-based care improvement interventions

To promote the adoption of evidence-based care improvement at the institutions, the Network established the SEAR-CCN Virtual Tumour Board and pathology external quality assurance systems.

SEAR-CCN Virtual Tumour Board (VTB)

Multidisciplinary team meetings are a critical part of the cancer care pathway as they bring together a diverse set of health-care professionals – including oncologists, radiotherapists, surgeons, pathologists, nurses and social workers – to pool their expertise and exchange the latest evidence to customize patient treatment plans. Tumour board meetings have proven to be instrumental in achieving accurate diagnoses, informed treatment options and better patient outcomes.[19] Such meetings are essential to make decisions about the clinical pathway of an individual patient in the context of available resources.

However, such a multidisciplinary approach is a great challenge in resource-limited settings due to the lack of requisite expertise. This was the case at many cancer treatment centres in the SE Asia Region, as revealed in the baseline institutional profile of the initial SEAR-CCN institutions.[16] In such circumstances, a virtual model of a tumour board could serve the same purpose. In a study on the use of digital resources and virtual tumour boards for multidisciplinary discussions on paediatric oncology cases in rural and other low-resource settings, it was shown that they led to improved clinical decision-making and increased the likelihood of appropriate treatment recommendations for paediatric cancer patients.[20]

In line with the aim of the SEAR-CCN to build a community of practice among institutions to implement evidence-based interventions for improving the quality of and access to care, the SEAR-CCN prioritized the organization of cross-country virtual multidisciplinary team meetings, also referred to as virtual tumour boards (VTBs), to support the management of difficult cases of childhood cancers at the institutions. Besides contributing to direct patient care, the establishment of the VTB also aimed at serving as a model for multidisciplinary care approach and promoting the adoption of evidence-based practices appropriate to the resource availability at the institution.

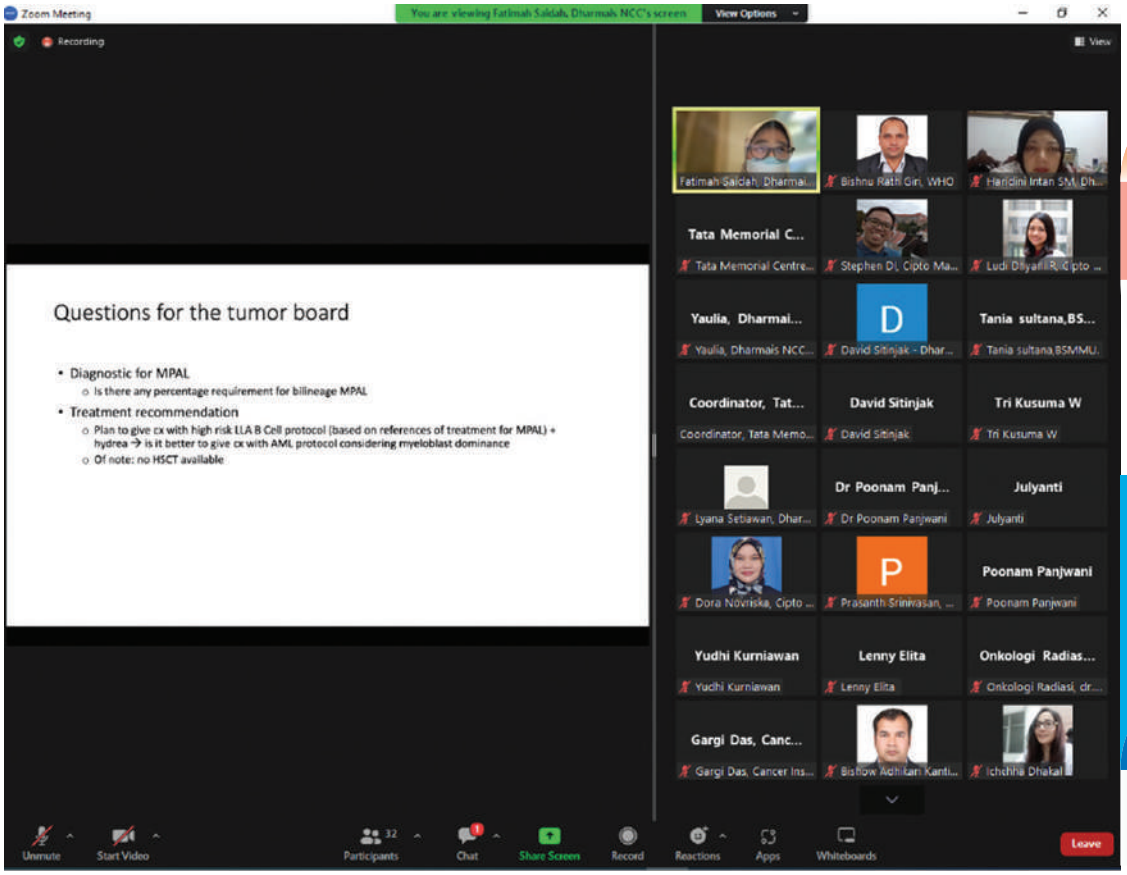
Initiation of SEAR-CCN activities during the peak of the COVID-19 pandemic was another reason why the VTB was preferred as a platform to discuss and recommend clinical approaches for the management of difficult cases from participating institutions across the Region. The task of coordinating the VTB was assigned to the Tata Memorial Hospital, based in Mumbai, India, which already had considerable experience in handling a similar platform – the National Cancer Grid for institutions in India.

The VTB sessions under the SEAR-CCN are held once every month (Fig. 5). Based on consensus among the participating institutions, the first Wednesday of every month was selected as the date for VTB meetings and to best suit the institutions in different time zones – 2:30 pm–4:00 pm (Indian Standard Time) was fixed. The cases to be discussed in the meeting are shared by the participating institutions at least two days in advance. In exceptional situations, late submissions are also included.

The submissions follow a specified template as a PowerPoint document, which includes anonymized clinical information, relevant images from pathology and radiology, and significant issues to be discussed. These cases are distributed to other participating institutions via email two days before the VTB meeting for preview. For the VTB meeting, participants include paediatric oncologists, surgical oncologists, radiation oncologists, radiologists, pathologists, nurses and other relevant professionals from the coordinating and other participating centres. Inputs on the cases are provided by experts participating from any centre, based on their knowledge and experience of managing similar cases.

A consensus plan on further management of patients is made, depending on the resources available at the presenting centre. The recommendations are sent via email to the treating centre

Fig. 5. Virtual Tumor Board meeting



subsequently. The experts participate in the VTB as part of their institutional work and there is no provision of additional payment for the participation.

Since its initiation in September 2021, 46 sessions of VTB meetings have been conducted till July 2025, with 35 centres from eight countries, including 17 SEAR-CCN member institutes, having participated in one or more sessions. Analysis of data on VTB meetings held till March 2025 showed that on average, 20 experts participated in each VTB, where 2–6 cases were discussed. Each VTB meeting was attended by experts from 4–6 institutions based in 3–5 countries.

The member institutes also facilitate sharing cases from institutes in their country, which are not formal members. Based on the participants' interest, special topics relating to pathology and local treatment planning are also presented in some sessions. Table 1 shows the summary of the utilization of VTBs by the institutes and the representation of countries.

Table 1. Summary of SEAR-CCN VTB operation (September 2021–March 2025)

Indicator	Numbers
Number of virtual tumor boards conducted	42
Number of institutions ever participated	35
Total and average (± SD) number of countries represented in each VTB	8, 4 (± 1.2)
Average number of institutions participating in each VTB meeting	6.0 (± 2.5)
Average number of cases discussed	3.5 (± 1.5)

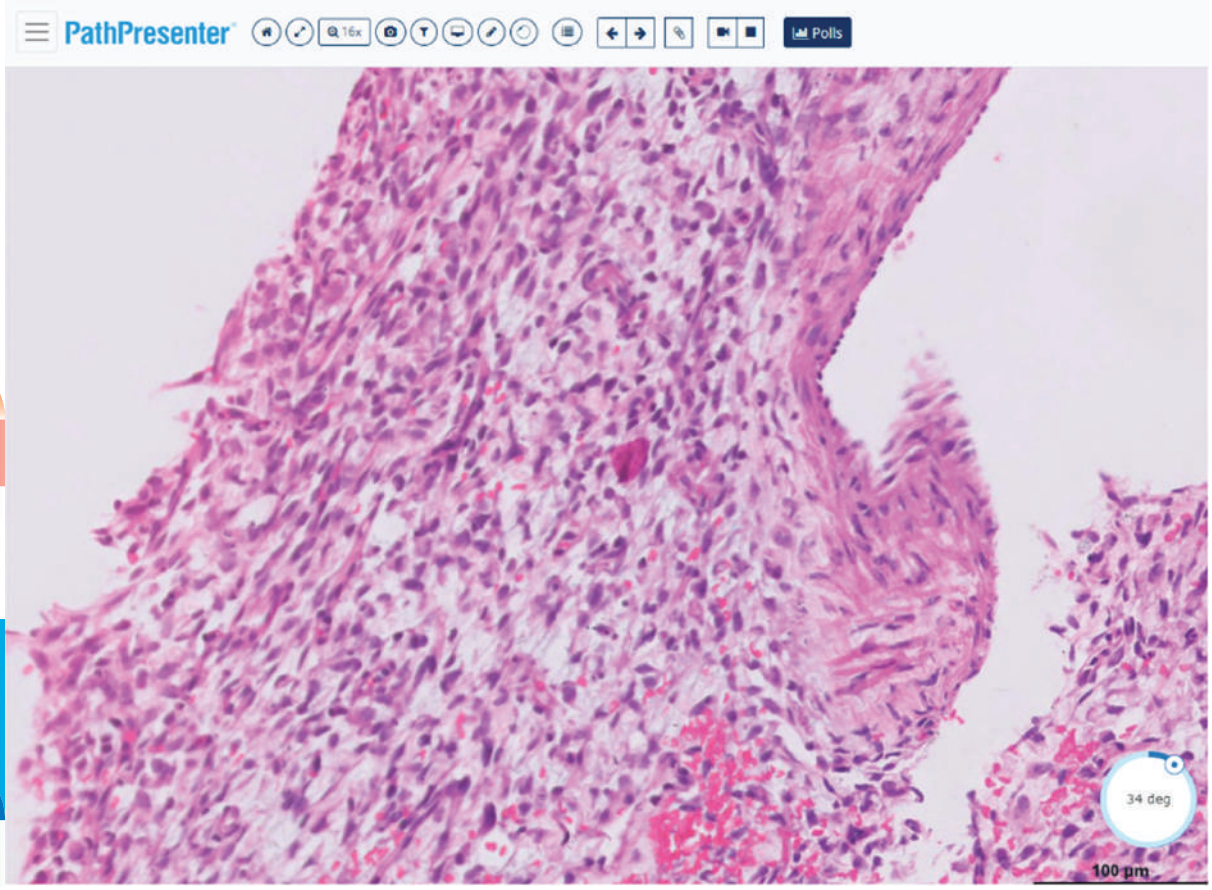
In a feedback survey conducted in 2023 among experts who had participated in one or more VTB sessions, participants said that they agreed with expert committee recommendations more than 90% of the time. These recommendations were implementable in most cases, except in situations of non-availability of drugs or lack of expertise for implementing them. Most centres have modified their existing protocols based on VTB recommendations. The VTB has served as part of the education and training for oncology fellows. Based on the experience of the SEAR-CCN VTB, 50% of the respondents planned to conduct a national or regional VTB in their country.

Pathology external quality assurance services

Following the observation that over 15% of the cases discussed in the SEAR-CCN VTB involved diagnostic dilemmas in pathology, the SEAR-CCN launched pathology external quality assurance services (eQAS) as a Network-wide activity in October 2023, with the dual objective of building capacity and ensuring quality in diagnostic services for childhood cancer.[17] Pathology quality assurance services help ensure that laboratory test results are reliable. Tata Memorial Hospital, with its experience in running an external quality assurance scheme for surgical pathology and molecular pathology under India's National Cancer Grid, was chosen to run a similar scheme for the SEAR-CCN. The eQAS activity is conducted virtually via a digital platform designed for such a purpose (Fig. 6).

The first eQAS cycle was launched in October 2023, with six institutions from four countries participating in it. By the end of 2024, three cycles of eQAS had been completed. Ten SEAR-CCN network member institutions from seven countries in the Region have participated in the programme.

Fig. 6. Histopathology slide seen on the online platform for pathology eQAS.



The participant institutes also included eight non-member institutions that participated through invitation (Annex 3). The pathology eQAS served a useful purpose in capacity-building and quality assurance of the participating institutions.

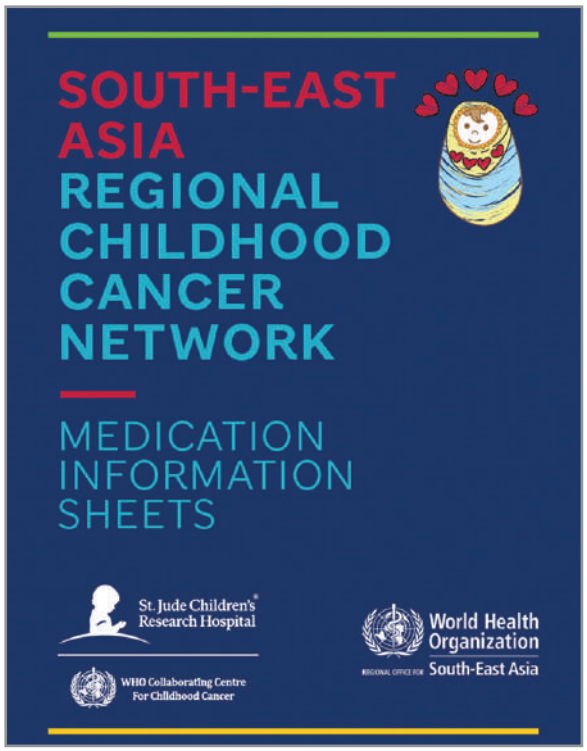
A review of the programme is being planned, involving interaction and feedback from the participants to identify ways to make it more effective. A formal assessment of the pathology eQAS will be planned subsequently to evaluate the impact on pathology diagnostic services and to document the usefulness of the approach.

Sharing expertise and resources to fill service gaps

Building and strengthening childhood cancer care networks within countries is one of the core activity areas for the SEAR-CCN. Facilitated by the SEAR-CCN, Nepal and Myanmar are operating a model of shared care network. Its objective is to shift the relatively less intense part of childhood cancer treatment from the tertiary-level principal treatment centre located centrally to secondary-/primary-level health facilities located peripherally, closer to patients' homes.[21,22]

In the shared care model, health facilities at the secondary/primary level require capacity-building and continued mentoring support from the principal treatment centre to deliver the less intense part of treatment, such as the maintenance phase treatment of acute lymphoblastic leukaemia, ensuring the same quality of care. While this model was being implemented in Myanmar with technical support from St. Jude Children's Research Hospital, providers at the shared care centres (also referred to as satellite centres in Myanmar) expressed the need for quick-reference guidance materials to support the management of children with cancer at the centres.

Fig. 7. Medication information sheets, one of the shared care centre resource guides in development by the SEAR-CCN



To strengthen these networks by making available accessible reference materials for use by providers at the point of care, the SEAR-CCN developed the shared care centre resource guides to facilitate the delivery of evidence-based quality care at the shared care sites.

The SEAR-CCN formed a Regional Working Group for this task, comprising expert representatives from the member institutions and some of the shared care centres to work further and develop regional resource guides. The Working Group had 34 members from 14 institutions from eight countries. The members included paediatric oncologists, radiation oncologists, paediatricians and nurses working in paediatric oncology. The Working Group collaborated with the partners and experts, including those from St. Jude Children's Research Hospital.

The priority topics to be included in the resource guides were determined via a prioritization exercise administered online for practitioners at the SEAR-CCN member institutions and the shared care

sites associated with them. Three different types of resource guides have been developed –medication information sheets, job aids/posters and a manual with chapters on the priority topics identified (Fig. 7). The manual chapters provide more detailed guidance on the topics referenced in medication sheets and posters that include managing oncology emergencies, safe handling of chemotherapy, managing extravasation, adverse effects of chemotherapy, febrile neutropenia and palliative care.

Building capacity of childhood cancer professionals

Nursing personnel should possess specialized skills in paediatric oncology to deliver good-quality care to address the complex needs of children with cancer. However, significant disparities exist not only in the nurse–patient ratio but also in insufficient nursing training between countries of different income categories.[23,24] In addition, participation of nursing staff in multidisciplinary care of cancer in children is less frequent in low- and middle-income countries.[25] The baseline institutional profiles of the initial SEAR-CCN member institutions also highlighted a similar trend in the Region, with only six out of 11 institutions having nurses who had received more than 40 hours of training in chemoadministration.[16]

Building capacity and developing training programmes for childhood cancer professionals is one of the core objectives of the SEAR-CCN. In line with this, the Network launched a virtual

Fig. 8. Hands-on training for paediatric oncology nurses

paediatric oncology nursing training programme, in partnership with Tata Memorial Hospital, a WHO collaborating centre. The training module was developed at both basic and advanced levels, covering topics relevant to paediatric oncology nursing, including nutrition, GICC index cancers and psychological counselling. So far, 170 nurses from 25 institutions (Annex 4), including 10 SEAR-CCN member institutions and 15 non-member institutions associated with them, have been provided with basic training, while 20 nurses have been trained at the advanced level. A two-week hands-on training was organized in July 2025 for 10 paediatric oncology nurses at the Network institutions (Fig. 8). These candidates were selected based on their potential to emerge as trainers at their respective institutions.

In addition, the SEAR-CCN facilitated hands-on training sessions for other categories of health-care professionals involved in care of cancers in children. These included a one-month hands-on training for four pharmacists from Yangon General Hospital, Myanmar, at Tata Memorial Hospital, Mumbai, in April 2023.

Fig. 9. Screengrab of a video from the training module on bone marrow aspiration and biopsy

To support standardization of some of the common clinical procedures in paediatric oncology, the SEAR-CCN has also developed training modules for physicians and nurses on paediatric procedural sedation, lumbar puncture and intrathecal chemotherapy, bone marrow aspiration and biopsy, and peripherally inserted central catheter insertion and care.

The development of the modules is led by the All India Institute of Medical Sciences; other SEAR-CCN member institutions were invited for review and inputs. The technical contents of the modules, including the learning objectives, manual, PowerPoint presentations and videos, have been finalized and are currently in the final designing stage (Fig. 9). These resources will be made available for use by all members of the SEAR-CCN for conducting training on those topics.

Fostering bilateral collaboration among member institutions

Another objective of the SEAR-CCN involves fostering collaborations among the member institutions. Horizontal collaborations between institutions have emerged notably with the progress in Network activities.

Fig. 10. **Training manual developed by the RSCM for training of staff at HNGV**

One such example involves a horizontal model of learning and training between a member institution in Jakarta, Indonesia, Cipto Magunkusumo Hospital (RSCM), and Hospital Nacional Guido Valadares (HNGV) in Timor-Leste to enable the initiation of basic childhood cancer services in the latter (Fig. 10). A similar collaboration is being established between Tata Medical Centre, Kolkata, India, and JDW National Referral Hospital in Thimphu, Bhutan.

In another example, Yangon Children's Hospital in Myanmar and Kanti Children's Hospital in Nepal have fostered a collaboration to exchange experiences in developing shared care centre networks within their respective countries.[20]

The learnings from similar initiatives in similar circumstances have benefitted both. Another collaboration between Kanti Children's Hospital and Tata Memorial Hospital, Mumbai, has resulted in onsite and accredited specialist training in paediatric oncology and paediatric oncology nursing.

In most of these collaborations, the resources have been mobilized by the institutions themselves.

4. Key initiatives for GICC implementation in the focus countries

Myanmar, Nepal and Sri Lanka were designated as focus countries when the activities under the GICC were initiated in the SE Asia Region in 2020. In May 2024, Bhutan was designated as the fourth focus country in the Region. Countries are designated as focus countries based on formal requests from the governments concerned.

This is followed by a WHO review of the commitment of the country to prioritizing childhood cancer in policies and systematically implementing programmes to improve services for children with cancer. Once designated, each focus country identifies its priorities and develops workplans and programmes, aligned with the **Cure*All*** framework, with technical support from WHO and partners, when needed.

The focus countries in the Region have strengthened multisectoral collaboration and mobilized resources. They are implementing priority actions with technical support from WHO, St. Jude Children's Research Hospital and other partners. The areas of work include addressing childhood cancer in policies and strategies, workforce capacity-building, standardizing treatment regimens, quality improvement in care, supporting access to childhood cancer medicines and advocacy, among others.[17]

The following sections describe some of these activities led by the countries in the Region.

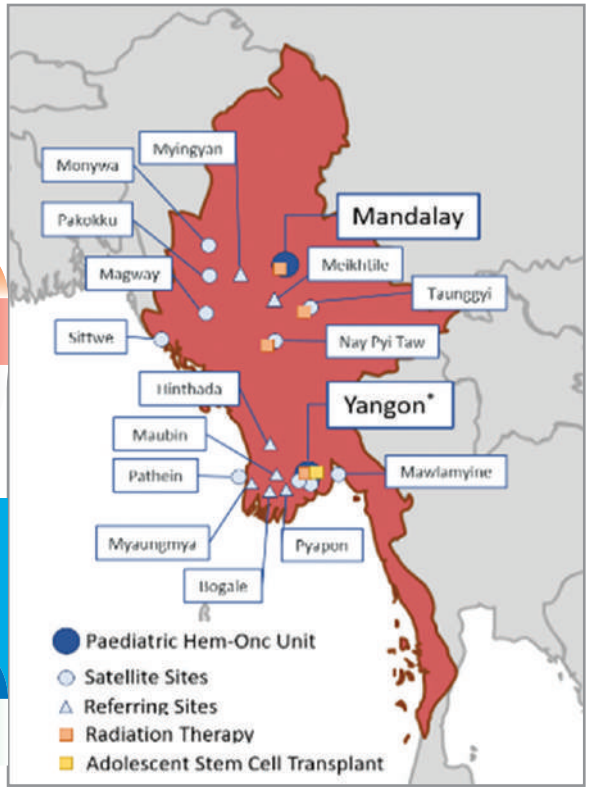
Myanmar: Establishing resilient childhood cancer services through the satellite centre network

In Myanmar, an estimated 1626 new cases of childhood cancer occurred in 2022.[9] These children are provided with cancer care services at two major hospitals – Yangon Children's Hospital (YCH) in Yangon and Mandalay Children's Hospital (MCH) in Chanayethazan township. Around 75% of the cases at YCH are referred from outside Yangon City. MCH has a similar scenario. The need to travel to a distant city for a child's cancer treatment significantly increases the burden on families, many of whom are economically disadvantaged. The data from YCH show that over 15% of newly diagnosed children with cancer abandon treatment within the first year of diagnosis.[20]

To mitigate the hardship families face during the treatment of children with cancer, YCH and MCH decided to collaborate with hospitals at lower levels of care in 2019. Five satellite sites were identified and linked to MCH, while 12 sites were connected to YCH (Fig. 11).

The satellite centres were selected by YCH and MCH based on their respective load of patients suitable for referral. The most important criterion for the selection of a satellite centre was the availability of a paediatrician who could attend to referred patients. Health-care staff at satellite centres were trained in basic paediatric oncology, including diseases, treatment regimens, handling of chemotherapy medicines and oncological emergencies. This was to build their capacity to deliver a relatively less intense part of the care for children receiving cancer treatment.

Fig. 11. **Distribution of 17 initial satellite centres in 2019 (Credit: Yangon Children's Hospital)**



To smoothen the referral process, a uniform mechanism for all satellite sites was developed and implemented. The details of a patient being referred, including the demographics, clinical details and treatment plans, are shared with the paediatrician at the satellite centre via telephone before the referral. In addition, patients are given a "patient book", which bears the case history, treatment plan, follow-up records, etc. Details are recorded by the principal as well as the satellite site in the same "patient book", which is carried by the family while visiting any of the treatment centres. Important information on diagnosis and treatment is extracted from the "patient book" and recorded on a computer at the principal treatment centre, when the patient visits it.

Whenever needed, the paediatrician at the satellite centre and the specialist at the principal treatment centre hold a virtual meeting for review and monitoring of the patient's progress. Short communications via phone and messaging platforms have helped strengthen clinical decision-making and boost health-care provider confidence. Patients being referred are also provided with transportation facilities. A mechanism has also been established for the supply of medicines to satellite centres.

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Service utilization at satellite site network hospitals increased gradually over time. During the COVID-19 pandemic, when the movement of people and transport was severely restricted, these hospitals saw a notable rise in utilization and proved instrumental in ensuring continuity of treatment. More patients opted for treatment at satellite hospitals during the difficult political situation that followed. Towards the end of 2022, WHO supported the recruitment of staff and supply of essential medicines for the continuity of childhood cancer services.

Network hospitals were encouraged to hold patient review meetings and resource guides were developed for satellite centres for point-of-care reference. Training of health-care providers continued virtually. The usefulness of these centres was evident in the aftermath of cyclone Mocha in May 2023, when a higher proportion of patients from the areas affected by the cyclone opted for treatment at satellite centres.

During the COVID-19 pandemic, the political crisis and the aftermath of cyclone Mocha, 115 health-care providers from 41 network hospitals in 15 states/regions were trained in paediatric oncology via virtual platforms. Over 900 patient reviews were conducted with 22 network hospitals from 13 states/regions between November 2022 and December 2023. Out of 491 new cases diagnosed at YCH in 2023, 32% of the patients were jointly treated through YCH and network hospitals.

The number of satellite centre network hospitals increased from 17 in 2019 to 41 in 2024. The innovative model of shared care with satellite site network hospitals has brought cancer service delivery closer to the homes of children with cancer and has helped provide resilience during crises. Continued technical support, mentoring from principal treatment centres and a reliable supply of essential medicines play a vital role in the success of this model.[26]

Nepal: Effective advocacy driving progress in childhood cancer

Advocacy and policy

In 2021, the Integrated Mission on Programme of Action for Cancer Therapy (imPACT) review of Nepal, conducted by WHO, International Atomic Energy Agency (IAEA) and International Agency for Research on Cancer (IARC), focused on childhood cancer, Nepal being one of the GICC focus countries. It was a first-of-its-kind for imPACT reviews globally. The review provided specific recommendations on childhood cancer, which formed a strong basis for many of the activities that followed.

Nepal organized a national workshop on "Accelerating Nepal Childhood Cancer Initiative" in June 2023 at Dhulikhel.[27] It was a pioneering gathering of a diverse group of stakeholders in Nepal, dedicated to addressing childhood cancer. The objective was to draw the attention of policy-makers and health officials to how childhood cancer services could be strengthened across the country. The workshop identified several priorities and stressed the need to formulate an informed and comprehensive childhood cancer plan, aligned with the National Cancer Strategy, along with an investment case.

The workshop was attended by the Minister of Health, Mohan B. Basnet, senior officials from the Ministry of Health, partners and civil society organizations. It adopted the Dhulikhel Declaration (Box 1) as a guiding framework, seeking transformative changes to elevate the quality of childhood cancer care and achieve improved outcomes and enhanced survival rates for children with cancer in Nepal. This was later officially endorsed by the Ministry of Health.

As an impact of the national consultation, childhood cancer received significant attention in the National Cancer Strategy (2024–2030), endorsed by the Ministry of Health in August 2024.

Box 1. Dhulikhel Declaration on improving childhood cancer care in Nepal

The Dhulikhel Declaration resolved to implement collaborative efforts to:

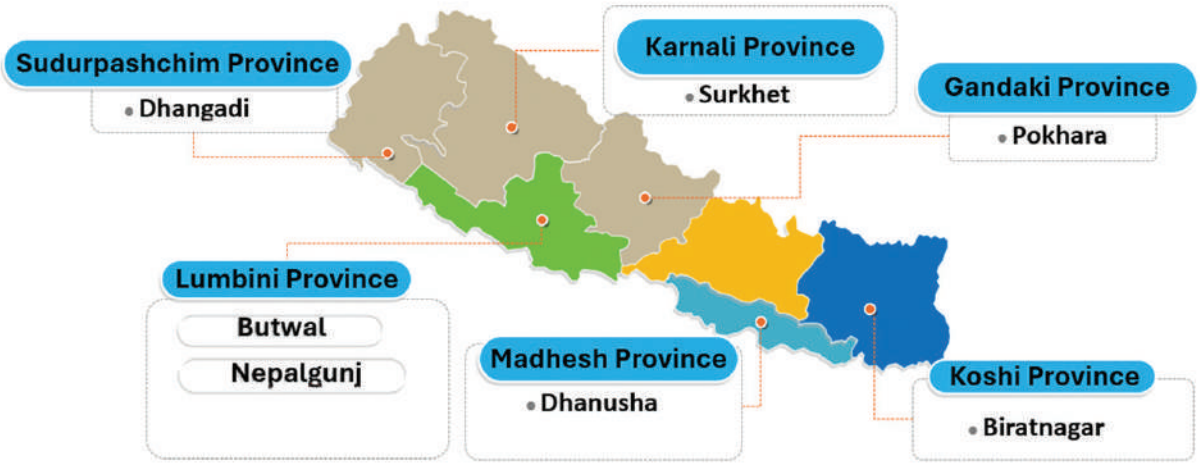
1. formulate an informed and comprehensive childhood cancer plan, aligned with the National Cancer Strategy, with an investment case, to inform relevant agencies and partners to ensure sustainable implementation of the plan;
2. define and apply minimum service standards for specialized childhood cancer centres and shared care centres, including expertise and human resources, diagnostic facilities, equipment and infrastructures, resource-stratified regimens, appropriate counselling and psychosocial support services, nutritional services, and systems for quality improvement, monitoring and evaluation;
3. ensure access to cytotoxic and essential medicines for childhood cancer treatment without financial burden on patients and their families, with a focus on optimizing supply chains, minimizing stockouts and ensuring quality control;
4. optimize support for families of children with cancer, including the cost of diagnosis, treatment and supportive care, and provide optimal support for indirect costs such as accommodation and travel expenses incurred during treatment, with the aim to reduce treatment refusal and abandonment, and enhance survival and quality of survival for children with cancer; and to make a provision for dedicated personnel to coordinate psychosocial support at specialized treatment centres;
5. strengthen diagnostic services for childhood cancer, including at a reference laboratory at the national level for high-end diagnostics;
6. expand and strengthen paediatric radiation therapy and optimize referral mechanisms across childhood cancer centres;
7. strengthen and expand paediatric palliative care services at all levels, including primary health care;
8. develop workforce training and fellowship programmes with accredited certification to ensure skill enhancement and development of expertise in childhood cancer within the country;
9. harmonize the management of childhood cancer across Nepal, including diagnosis, treatment and supportive care, at both specialized childhood cancer centres and shared care centres;
10. establish networking among institutions to facilitate the efforts on harmonizing the management and sharing of patient care across the network of centres of excellence in different disciplines;
11. establish a single, common hospital-based childhood cancer registry with mandatory participation from all childhood cancer treatment centres enforced by the ministry, with the provision of dedicated data managers at each specialized centre, supported by regular training and clear policies for retention;
12. strengthen the capacity of general and basic health care facilities in early detection and referral of childhood cancers, and include Early Warning Signs and Symptoms (EWSS) in national guidelines for treatment of childhood illnesses;
13. incorporate the perspectives of civil society organizations and patients/families into structured planning and consultation processes for childhood cancer; and
14. ensure that by 2030, every child developing cancer in the country is diagnosed and started on timely quality treatment until a defined outcome is met.

Shared care network for improving access

An estimated 1300 new cases of childhood cancer occur in Nepal every year. However, only around a third of them receive treatment, resulting in much lower cure rates, compared with those in high-income countries. Patients and their families from different parts of the country are forced to travel to tertiary care centres located in urban areas. The health facilities at the zonal or provincial level lack the capacity for independent management of childhood cancer. The shared care centre model for childhood cancer in Nepal involves organized networking of tertiary treatment centres with zonal or provincial health facilities (Fig. 12).[21] To begin with, the model helped in delivering maintenance phase treatment for acute lymphoblastic leukaemia, which is the most common form of childhood cancer in Nepal.

At the centre of shared care delivery is Kanti Children's Hospital (KCH) in Kathmandu. The tertiary care centre is equipped with facilities for childhood cancer care and serves patients from all over Nepal. It has networked with seven other hospitals for shared care service delivery, with technical support from WHO, St. Jude Children's Research Hospital and World Child Cancer, a global charity organization.

Fig. 12. Shared care centre network in Nepal



As a first step, the paediatric oncology team of KCH developed a curriculum and training material for a short course in December 2021. The two-day training for health-care professionals, held onsite at the shared care centre, covered topics related to management of children with acute leukaemia, with a focus on maintenance phase treatment and emergencies. The training programme participants included paediatricians, general oncology physicians, general physicians and nurses.

The training was followed by the initiation of maintenance phase treatment – monthly pulses of vincristine and dexamethasone – for children with acute lymphoblastic leukaemia and non-complicated follow-up visits at the shared care centres. Initially, individual case-based telephonic consultation was provided by the KCH team, as needed. Families were required to visit KCH just once in three months for receiving intrathecal chemotherapy, instead of the usual monthly visits earlier. This drastically reduced the cost of travel, loss of wages or absence from work for parents and school absenteeism for children.

National guidelines for Wilms tumour treatment

In an effort to standardize and establish uniformity in treatment regimens, Nepal has started with Wilms tumour as the initial focus among the GICC index cancers.[28] Experts from various specialities from all major centres managing childhood cancer have worked together to develop a set of guidelines for diagnosis and treatment of children with Wilms tumour. The guidelines have also identified key indicators for monitoring compliance. WHO and St. Jude Children's Research Hospital provided the technical support during this exercise.

Improving access to childhood cancer medicines

Nepal has been selected as one of the six pilot countries to participate in the Global Platform for Access to Childhood Cancer Medicines (GPACCM).[29] The Platform was announced by WHO and St. Jude Children's Research Hospital in December 2021 to address problems concerning access to medicines for childhood cancer. In the initial phase, the Platform will support countries with 35 types of childhood cancer medicines.

After a series of preparatory works, including assessment of minimum standards, addressing the critical gaps in supply chain management, capacity-building and development of the GPACCM implementation plan, Nepal has begun receiving the medicines from February 2025.[30] Nepal will receive the medicines free of cost for five years through the GPACCM. The distribution of medicines will be expanded to the shared care centre hospitals on a need basis in the future.

With the advocacy carried out through the GICC, the resulting political commitment, the necessary policy changes and the support from the GPACCM, in November 2024, Nepal declared free treatment for children with cancer at public health facilities.[31]

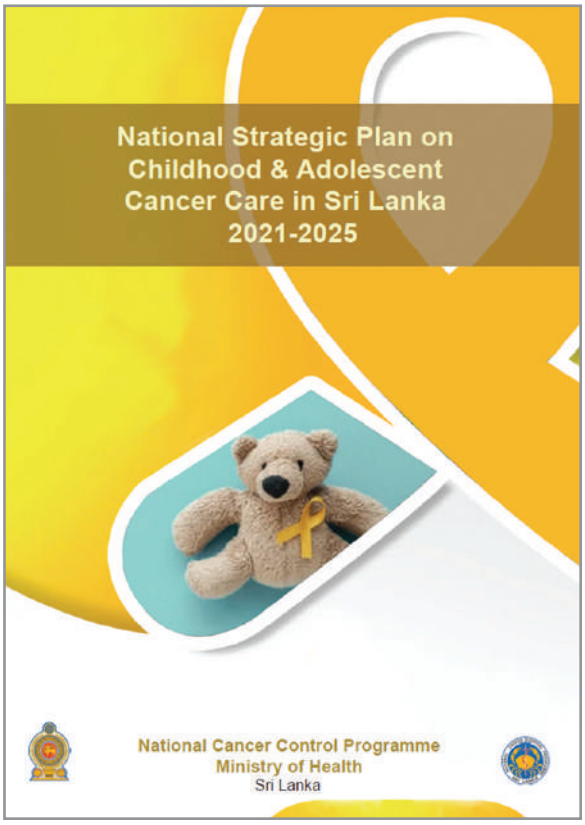
In addition, effective partnership with various civil society organizations has enabled delivery of holistic care to children and families, including psychosocial and family support and survivorship care. Some other works in progress include strengthening hospital-based childhood cancer registry and developing paediatric palliative care training manual and guidelines for Burkitt lymphoma.

Sri Lanka: Leading the way in childhood cancer through its National Strategic Plan

As a GICC focus country in the SE Asia Region, Sri Lanka developed the National Strategic Plan (NSP) on Childhood and Adolescent Cancer Care in Sri Lanka 2021–2025 as an extension of the National Strategic Plan on Prevention and Control of Cancers in Sri Lanka 2021–2024 (Fig. 13). The objective of the National Plan on childhood cancers is to serve as the main guiding document for Sri Lanka to achieve the GICC targets.[32]

The goal of the Plan is to achieve high survival rates through timely diagnosis and effective treatment and follow-up care by providing continuum of care to all childhood and adolescent cancer patients in an equitable manner. The Plan has identified specific action areas: improving treatment infrastructure; networking of cancer treatment centres; training health-care professionals in a multidisciplinary approach for handling paediatric cancers, including the constitution of tumour boards and multidisciplinary team-based decision-making; strengthening histopathology services; establishing national centres of excellence; development of a National Essential Medicines List for childhood cancers and providing access to essential medicines; and promotion of cancer research.

Figure 13: **National Strategic Plan on childhood and adolescent cancer, Sri Lanka**



The NSP also identified the need to design an information system, specifically for childhood cancers. A dedicated NSP has helped Sri Lanka implement programmes and activities for improving childhood cancer care in a coordinated and effective manner, including development of clinical practice guidelines and training modules for nurses.[33]

Establishment of hospital-based cancer registry

Among the notable initiatives to improve childhood cancer in Sri Lanka is the establishment of a hospital-based childhood cancer registry at the National Cancer Institute. The registry uses the SJCARES platform developed by St. Jude Children's Research Hospital, specifically designed to suit the needs of low- and middle-income countries. The data from the National Cancer Institute and other sources were used to publish the first report on childhood cancer registry in Sri Lanka.[34] Sri Lanka is the only country in the Region to publish a cancer registry report specifically meant for childhood cancer.

Establishment of an antibiotic stewardship programme

Infections are a major cause of morbidity and mortality among children undergoing treatment for cancers. Irrational use of antibiotics further complicates the matter by promoting antimicrobial resistance (AMR). Therefore, steps were taken to develop an antibiotics policy for optimizing their use and implementing an antibiotics stewardship programme at the paediatric oncology unit of the National Cancer Institute.

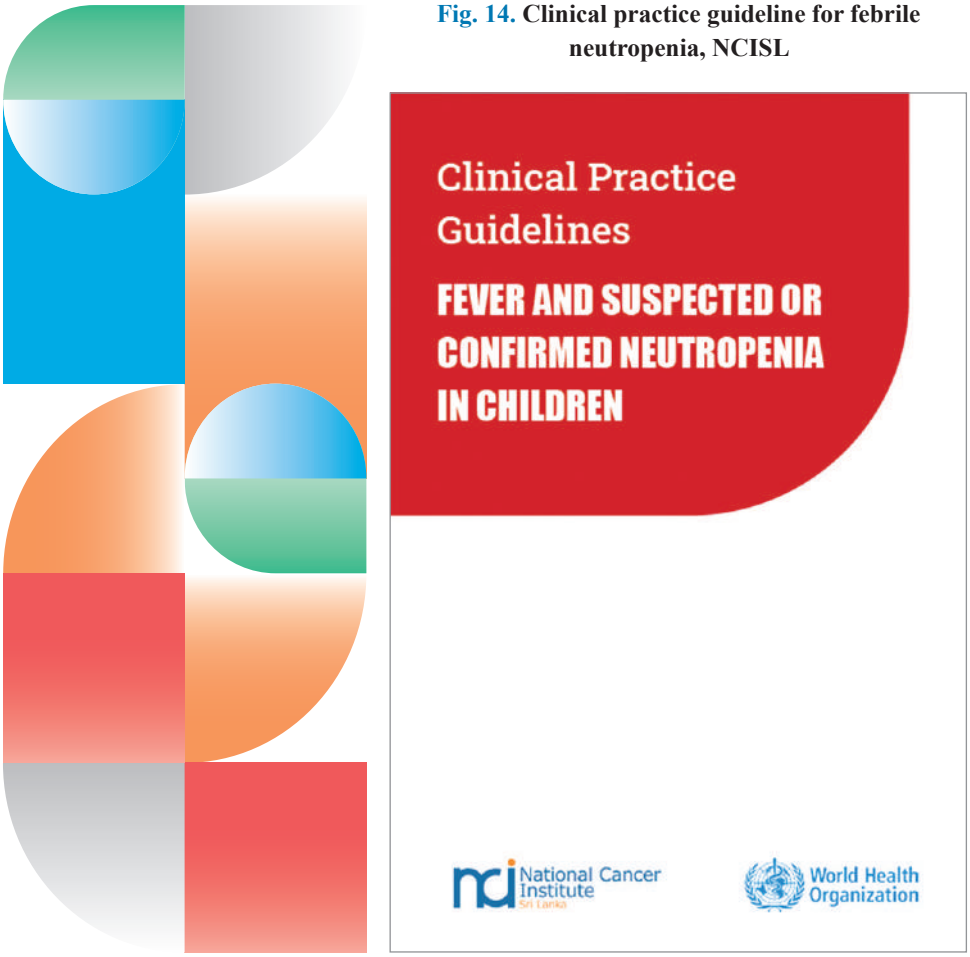


Fig. 14. **Clinical practice guideline for febrile neutropenia, NCISL**

An initial stakeholder meeting within the institute concluded that a preliminary audit was necessary to gain more knowledge on the status of antibiotic usage. This revealed that about 90% of the antimicrobial prescriptions were empirical and another 7% were prophylactic. This led to the development of clinical practice guidelines for the initial assessment and management, ongoing assessment and antimicrobial agents of choice, including dose and frequency for patients (Fig. 14). Drug charts were amended to include a separate section for antimicrobials.

This was shared formally through interactive training meetings and use of posters at strategic locations in the clinic and wards.

Development of pain-free procedure protocol

Children with cancer undergo frequent invasive procedures, such as bone marrow sampling and lumbar puncture, that cause a significant amount of pain. Having a standard protocol for reducing or abolishing pain becomes a crucial part of good-quality care for childhood cancer. To achieve this, the National Cancer Institute Sri Lanka (NCISL) team proposed developing an institution-specific, pain-free procedure protocol.

This began with a literature review, followed by a series of technical meetings, with the participation of paediatric oncologists and anaesthesiologists, among others. The draft thus prepared was circulated among relevant specialities within the hospital and the feedback was incorporated into the final version of the protocol.

The staff involved in implementing the protocol received training, with over 100 participants completing the training over two sessions. Pre- and post-training assessments showed a 20% improvement in participants' knowledge from baseline.

This was also accompanied by upgrades to procedure rooms, including equipment, to facilitate pain-free procedures and create a more child-friendly environment.

Guidelines for pathological diagnosis of nephroblastoma and retinoblastoma

Sri Lanka has developed guidelines on the pathological diagnosis of two GICC index cancers – nephroblastoma and retinoblastoma – as part of the effort to improve the overall quality of care for childhood cancer. The guidelines cover specimen collection, template requisition form, SOP for transportation of specimen, specimen handling and processing at laboratories, examination and interpretation, and reporting. The guidelines are expected to support establishment of uniform standards for diagnosis and lead to reliable staging, treatment planning and prognostication of Wilms tumour and retinoblastoma.

Improving access to childhood cancer medicines

Sri Lanka has been facing frequent stockouts of medicines for childhood cancers, with reportedly only less than half of the antineoplastic agents on the WHO Model Essential Medicines List being consistently available.[35] Noting the progress made in implementing the GICC and the favourable policy framework in the country, Sri Lanka has been selected among the countries participating in the Global Platform for Access to Childhood Cancer Medicines (GPACCM) in the second phase. Initial preparatory works for the participation, including minimum standards assessment, have begun. Participation in the GPACCM will help ensure consistent availability of quality-assured medicines for childhood cancers, while relieving the funds that can be used for availing other supportive care medicines.

5. Enablers of the Cure*All* framework facilitating the progress in the GICC in the SE Asia Region

The GICC activities in the SE Asia Region are aligned well with the four pillars of the **Cure*All*** framework – centres of excellence, universal health coverage, regimens for management, and evaluation and monitoring. They have strengthened the enablers of the framework – advocacy, leveraging financing and linked governance.

Advocacy helps create an environment favourable to prioritizing childhood cancer in policy and programmes. Drawing from the experience of the three initial focus countries in the SE Asia Region, advocacy efforts have been crucial in generating the political will for addressing childhood

The paediatric oncology team at the National Cancer Institute, Sri Lanka

cancer with significant emphasis on policies. The advocacy efforts, including national stakeholders' consultation, resulted in Member countries prioritizing childhood cancer and formulating policies well aligned with other health policies.

A clearly defined **governance** structure **linked to policy** greatly facilitated progress in childhood cancer in Nepal and Sri Lanka. In Nepal, a steering committee to guide childhood cancer programmes has been formed; it is chaired by the Secretary of the Ministry of Health. A Technical Working Group has also been established to oversee the technical matters. In Sri Lanka, the Directorate of National Cancer Control Programme governs childhood cancer activities in the country.

The combined impact of the three interconnected enablers has generated political will and facilitated **leveraging of financial resources** from government, civil society and partners, as evident in focus countries, including Nepal and Sri Lanka. This has created the means and strong momentum for numerous activities on improving the quality of cancer care and access to it.

6. Considerations for sustaining the achievements

Over the past four years, the progress in implementing the GICC in the WHO SE Asia Region has been substantial. This was noted when the GICC focus countries presented their progress and childhood cancer received significant attention in the South-East Asia Regional Workshop on scaling up services for cancer and implementing the South-East Asia Regional Cancer Grid.[8] Influenced by the progress in the initial focus countries, other countries, including Bangladesh and Indonesia, have expressed interest in increasing the level of GICC engagement with designation as a focus country.

Thus, some of the approaches in GICC implementation can serve as a model for other countries and a pathfinder for cancer control in general. However, there are certain areas, including those identified as priority for the SEAR-CCN, requiring more effort for sustaining the achievements and accelerating the progress. The interventions should be instituted through a comprehensive and multifaceted approach involving various stakeholders, strategies and resources.

Some of the important considerations for ensuring sustained and accelerated progress include:

- Countries with policies in place to address childhood cancer should ensure sustained political will for implementation. For countries that are yet to adequately address childhood cancer in policies, there should be strong advocacy efforts to develop childhood cancer policies aligned with other cancer and health-related policies. The priorities in policies should evolve with progress over the years.
- Formal engagement with the GICC provides opportunities for increased collaboration, thus countries in different levels of engagement should be encouraged for designation as a GICC focus country.
- Principal treatment centres should be equitably distributed across the country. They should be equipped with adequate expertise and human resources, treatment regimens, systems and infrastructure commensurate with the patient load as centres of excellences.
- Governments need to allocate adequate financial resources to support the establishment of centres of excellences. Human resource development and deployment require significant time, investments and planning.
- Shared care networks can be expanded to cover wider geographical access to care. They should provide quality-assured treatment and have well-coordinated referral mechanisms linked to principal treatment centres.
- As identified among the core activity areas for the SEAR-CCN, childhood cancer networks should be established within the countries for sharing resources and technical expertise. Similarly, bilateral and multilateral collaborative research activities need to be carried out to generate evidence for intervention in the local context.

- Childhood cancer information systems should be strengthened. The foremost priority lies in strengthening hospital-based cancer registries.
- Innovative mechanisms are needed for financing the cost of care when the publicly funded mechanisms are inadequate to support the treatment of a child for the entire duration. Locally relevant innovative mechanisms should be explored.

Childhood cancer survivors participating in a cultural performance at the All India Institute of Medical Sciences, New Delhi, India

Annex 1

Availability of medicines and supplies for childhood cancer care, as reported by countries in 2021[11]

	Bangladesh	Bhutan	India	Indonesia	Maldives	Myanmar	Nepal	Sri Lanka	Thailand	Timor-Leste
National EMLc	No	No	Yes	Yes	Yes	Yes	No	No	Yes	No
Chemotherapy medicines										
6-mercaptopurine	Always available	Not available	Mostly available	Frequently unavailable	Frequently unavailable	Mostly available	Mostly available	Mostly available	Frequently unavailable	Frequently unavailable
Actinomycin D	Mostly available	Not available	Mostly available	Mostly available	Not available	Mostly available	Mostly available	Mostly available	Mostly available	Frequently unavailable
Carboplatin	Always available	Frequently unavailable	Mostly available	Always available	Not available	Mostly available	Mostly available	Mostly available	Always available	Frequently unavailable
Cyclophosphamide	Always available	Mostly available	Mostly available	Always available	Not available	Mostly available	Mostly available	Mostly available	Always available	Not available
Cytarabine	Mostly available	Not available	Mostly available	Always available	Not available	Mostly available	Mostly available	Mostly available	Always available	Not available
Dexamethasone	Always available	Always available	Mostly available	Always available	Not available	Always available	Mostly available	Always available	Always available	Always available
Doxorubicin	Always available	Always available	Mostly available	Mostly available	Mostly available	Mostly available	Mostly available	Mostly available	Always available	Frequently unavailable
Etoposide	Always available	Always available	Mostly available	Always available	Not available	Mostly available	Mostly available	Mostly available	Always available	Frequently unavailable
L-asparaginase	Not available	Not available	Mostly available	Always available	Not available	Mostly available	Mostly available	Mostly available	Always available	Not available
Tab. Methotrexate	Always available	Always available	Mostly available	Frequently unavailable	Always available	Mostly available	Mostly available	Mostly available	Always available	Always available

	Bangladesh	Bhutan	India	Indonesia	Maldives	Myanmar	Nepal	Sri Lanka	Thailand	Timor-Leste
Inj. Methotrexate	Always available	Frequently unavailable	Mostly available	Always available	Not available	Mostly available	Mostly available	Mostly available	Always available	Mostly available
Vincristine	Always available	Mostly available	Mostly available	Always available	Not available	Mostly available	Mostly available	Mostly available	Always available	Frequently unavailable
Non-chemotherapy medicines and supplies										
Allopurinol	Always available	Always available	Mostly available	Always available	Always available	Mostly available	Always available	Mostly available	Always available	Always available
Tab. Morphine	Mostly available	Always available	Mostly available	Mostly available	Always available	Mostly available	Mostly available	Mostly available	Always available	Frequently unavailable
Ondansetron	Always available	Always available	Always available	Always available	Always available	Mostly available	Mostly available	Mostly available	Always available	Mostly available
LP needle	Always available	Always available	Mostly available	Always available	Not available	Always available	Always available	Always available	Mostly available	Mostly available
BM aspiration and biopsy needle	Always available	Always available	Mostly available	Always available	Not available	Mostly available	Mostly available	Always available	Mostly available	Mostly available

*EMLc: Essential Medicines List for children.

Annex 2

List of SEAR-CCN member institutions

Country	Institution
Bangladesh	National Institute of Cancer Research and Hospital, Dhaka, Bangladesh
Bhutan	Jigme Dorji Wangchuk National Referral Hospital, Bhutan
India	Lady Hardinge Medical College and Associated SSK and KSC Hospitals, New Delhi, India
	Tata Memorial Center, Mumbai, India
	Kidwai Medical Institute of Oncology, Bengaluru, India
	Dr B Barooah Cancer Institute, Guwahati, India
	All India Institute of Medical Sciences, New Delhi
	Tata Medical Center – Cancer Hospital and Research Centre, Kolkata
	Christian Medical College, Vellore, Tamil Nadu
	Cancer Institute (WIA), Adyar, Chennai
Indonesia	Dharmais Cancer Hospital, Jakarta, Indonesia
	Dr Cipto Mangunkusumo General Hospital, Jakarta, Indonesia
	Indonesian Paediatrician Association, Indonesia
Maldives	Hulhumale Hospital, Maldives
Myanmar	Yangon Children's Hospital, Yangon, Myanmar
	Mandaley Children's Hospital, Myanmar
Nepal	Kanti Children's Hospital, Kathmandu, Nepal
	BP Koirala Memorial Cancer Hospital, Chitwan
Sri Lanka	National Cancer Institute, Colombo, Sri Lanka
Thailand	Phramongkutklao Hospital and College of Medicine, Bangkok, Thailand
Timor-Leste	Hospital Nacional Guido Valadares, Dilli, Timor-Leste

Annex 3

List of institutions enrolled for pathology external quality assurance services till the fourth cycle

Sr. No.	Name of the institution and country
01	Central Regional Referral Hospital, Bhutan
02	Mongar Regional Referral Hospital, Bhutan
03	Dr B Borooah Cancer Institute, Guwahati, Assam, India
04	Hospital National Guido Valadares, Dili, Timor-Leste
05	Indira Gandhi Memorial Hospital, Maldives
06	Patan Academy of Health Sciences, Nepal
07	Institute of Medicine, Nepal
08	Jigme Dorji Wangchuck National Referral Hospital, Bhutan
09	Kanti Children's Hospital, Nepal
10	National Institute of Cancer Research and Hospital, Bangladesh
11	Lady Ridgeway Hospital for Children, Sri lanka
12	Yangon Children's Hospital, Myanmar
13	Faculty of Medicine University of Kelaniya Ragama, Sri Lanka
14	National Hospital, Kandy, Sri Lanka
15	Faculty of Medicine, University of Colombo, Sri Lanka
16	National Cancer Institute, Maharagama, Sri Lanka
17	Dr Cipto Mangunkusumo General Hospital, Indonesia
18	Colombo South Teaching Hospital, Kalubowila, Sri Lanka

Annex 4

List of institutions participating in nursing training (virtual)

Sr. No	Name the institution and country
01	Apeksha Hospital – National Cancer Institute, Sri Lanka
02	Bangabandhu Sheikh Mujib Medical University, Bangladesh
03	Dhading District Hospital, Nepal
04	Dr M R Khan Shisu Hospital and Institute of Child Health, Bangladesh
05	Maharaj Nakhon Chiangmai Hospital, Chiangmai, Thailand
06	Queen Sirikit National Institute of Child Health, Thailand
07	Ramathibodi Hospital, Mahidol University, Bangkok, Thailand
08	Samitivej Children Hospital, Thailand
09	Siriraj Piyamaharajkarun Hospital, Thailand
10	Thammasat University Hospital, Thailand
11	Vajira Hospital, Thailand
12	Wocare Foundation/Indonesia
13	Yangon Children Hospital, Myanmar
14	Dr Cipto Mangunkusumo Hospital, Jakarta, Indonesia
15	Jigme Dorji Wangchuck National Referral Hospital, Bhutan
16	Patan Academy of Health Sciences, Nepal
17	RSUP Prof. Dr R.D. Kandou Manado, Indonesia
18	Kidwai Memorial Institute of Oncology, Bengaluru, India
19	RSCM Cipto Mangunkusumo General Hospital, Indonesia
20	Tangerang District Hospital, Banten, Indonesia
21	Dr Kariadi Hospital, Semarang, Central Java Province, Indonesia
22	National Institute of Cancer Research and Hospital, Mohakhali, Dhaka
23	Hulhumale Hospital, Huvandhumaa Hingun, Mal , Maldives
24	BP Koirala Memorial Cancer Hospital, Chitwan, Nepal
25	Pharmongkutklao Hospital and College of Medicine, Thailand

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